

# YUKI OKAMOTO

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## RESEARCH INTEREST

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- Deep generative models
- Probabilistic modeling
- Bayesian inference
- Statistical machine learning
- Signal and image processing
- Inverse problems

## EDUCATION

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### KTH Royal Institute of Technology

Master of Science in Applied and Computational Mathematics

Aug 2024 - Jun 2026

*Stockholm, Sweden*

### The University of Tokyo

Bachelor of Engineering in Systems Innovation

Exchange studies at KTH during Aug 2022 - Jun 2023

Apr 2020 - Mar 2024

*Tokyo, Japan*

## RESEARCH EXPERIENCE

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### KTH Royal Institute of Technology

Master's Thesis Student at Division of Probability, Mathematical Physics and Statistics

Jan 2026 - Jun 2026

*Stockholm, Sweden*

Supervisor: Asst. Prof. Soon Hoe Lim

Thesis Title: Planning as Conditional Generative Sampling: Belief- and Reward-Conditioned Diffusion Planners for Informative Path Planning

- Developed a conditional generative framework utilizing diffusion models (DDPMs) to sample high-reward trajectories for autonomous underwater vehicles.
- Evaluated the model's out-of-distribution generalization, proving that feasible OOD conditioning allows the planner to synthesize novel behaviors absent from the training demonstrations.

### KTH Royal Institute of Technology

Research Assistant at Medical Imaging Physics Group

Oct 2025 - Current

*Stockholm, Sweden*

Supervisor: Prof. Mats Danielsson, Assoc. Prof. Mats Persson, Per Lundhammar

- Designed and implemented a permutation-equivariant deep learning model using Set Transformer to infer temporal ordering of unordered photon detection data in SPECT imaging.
- Reduced inference complexity from factorial time to a single forward pass while achieving competitive ordering accuracy for short chain lengths against a maximum likelihood baseline.

### National Ocean Policy Secretariat, Cabinet Office, Government of Japan

Research Assistant

Mar 2024 - Aug 2024

*Tokyo, Japan*

Supervisor: Prof. Toru Sato, Toshiro Suzuki, Takamasa Nakamura

- Developed a stochastic risk assessment model using Bayesian Monte Carlo simulations to quantify the frequency and magnitude of global mineral resource supply disruptions.
- Evaluated the economic security value of domestic mineral resource development by modeling the avoided production losses enabled by strategic stockpiling.

### The University of Tokyo

Bachelor's Thesis Student at Laboratory of Environmental Modeling and Synthesis

Mar 2023 - Mar 2024

*Tokyo, Japan*

Supervisor: Prof. Toru Sato

Thesis Title: Proposal and cost evaluation of observation point placement method for underwater monitoring of sub-seabed CCS

- Developed a cost-efficient monitoring framework for offshore carbon capture and storage (CCS).
- Performed large-scale numerical simulations to analyze environmental impacts of CO<sub>2</sub> leakage.
- Designed and implemented a modified greedy solution for optimal coverage problems.

## PUBLICATIONS

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- [1] **Y. Okamoto**, T. Sato, and S. Kanao, “Cost assessment of optimal seawater monitoring for unexpected CO<sub>2</sub> leakage from an offshore reservoir,” *International Journal of Greenhouse Gas Control*, vol. 146, p. 104455, 2025, ISSN: 1750-5836. DOI: <https://doi.org/10.1016/j.ijggc.2025.104455>.

## PRESENTATIONS

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### Conference Oral Presentations (underline: presenter)

- [1] **Y. Okamoto** and T. Sato, “Proposal and cost evaluation of observation point placement method for underwater monitoring of sub-seabed CCS,” presented at 17th Greenhouse Gas Control Technologies, Calgary, Canada, October 20-24, 2024.
- [2] **Y. Okamoto** and T. Sato, “Proposal and cost evaluation of observation point placement method for underwater monitoring of sub-seabed CCS,” presented at The Japan Society of Naval Architects and Ocean Engineers 2024 Annual Spring Meeting, Kanazawa, Japan, May 27-28, 2024.

### Seminars

- [3] **Y. Okamoto**, “Diffusion Planner for Underwater CO<sub>2</sub> Monitoring”, presented at KTH Climate Action Centre, Stockholm, Sweden, May 26, 2026.
- [4] **Y. Okamoto**, “Proposal and cost evaluation of observation point placement method for underwater monitoring of sub-seabed CCS,” presented at Imperial College London (UK), University of Bergen (Norway), and Plymouth Marine Laboratory (UK), February 10-18, 2024.

## RESEARCH GRANTS & FELLOWSHIPS

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### Environmental Research Grant

Miljöfonden, Swedish Engineers Association

Awarded for a thesis project focused on environmental sustainability.

Dec 2025

50,000 SEK

### Travel Grant

Musha Shugyo Program, The University of Tokyo

Funded research presentations at the University of Bergen, Imperial College London, and Plymouth Marine Laboratory.

Feb 2024

450,000 JPY

### Graduate Fellowship

Takenaka Scholarship Foundation

Highly selective national scholarship (max. 3 recipients annually) covering two years of graduate studies, extendable for doctoral studies.

Nov 2023

10,000,000 JPY

### Study Abroad Fellowship

Gyomu Super Japan Dream Foundation

Competitive scholarship awarded to support exchange studies at KTH Royal Institute of Technology.

Jul 2022

1,350,000 JPY

## SKILLS

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### Programming

Python, Git, MATLAB

### Tools & Frameworks

PyTorch, GPyTorch, Hydra, NumPy, Pandas, Matplotlib, Scikit-learn

### Language Skills

Japanese (native), English (professional), Swedish (advanced)

### Soft Skills

Leadership, Project management, Problem solving, Team collaboration

## EXTRACURRICULAR ACTIVITIES

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### TEDxUTokyo

Board Member

Nov 2021 - Jun 2022

Tokyo, Japan

- Co-organized a TEDx conference at the University of Tokyo with 400 participants.
- Managed logistics, coordinated with stakeholders, and secured over 10 enterprise sponsorships.

## REFEREES

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**Prof. Toru Sato**, *Bachelor's thesis supervisor*  
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The University of Tokyo  
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**Assoc. Prof. Mats Persson**, *RA supervisor*  
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KTH Royal Institute of Technology  
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**Prof. Mats Danielsson**, *RA supervisor*  
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**Asst. Prof. Soon Hoe Lim**, *Master's thesis supervisor*  
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